



National University of Modern Languages

Assignment No 1

Name:	Tayyaba Ejaz (2751)
Course:	Computer Graphics
Class:	BSCS VII-A
Due Date:	28th-March, 2025
Submitted To:	Mam Sabina Irum

Table of Contents

Task: Apply various concepts & related to 2d and 3d Computer Graphics.	3
1 You can see three images below, picture 1 , picture 2 and picture 3 . you have to choose only 1 image and create it by applying the basic concepts studied in class.....	3
2 Draw basic objects using lines, point, polygons etc. (complete drawing	3
3 Apply different transformations using the object properties to view a final scene without animations.....	3
4 Bonus points for those who will create accurate images	3
5 Concepts that are not studied till yet in the class will be marked 0.....	3
1.1 Code.....	3
1.2 Output	13
1.3 Description.....	13

Task: Apply various concepts & related to 2d and 3d Computer Graphics.

- 1 You can see three images below, picture 1 , picture 2 and picture 3 . you have to choose only 1 image and create it by applying the basic concepts studied in class.
- 2 Draw basic objects using lines, point, polygons etc. (complete drawing
- 3 Apply different transformations using the object properties to view a final scene without animations.
- 4 Bonus points for those who will create accurate images
- 5 Concepts that are not studied till yet in the class will be marked 0.



1.1 Code

```
#include <iostream>
#include <Windows.h>
#include <gl/GL.h>
#include <gl/glut.h>

void MyInit() {
    glClearColor(0.6, 0.8, 1.0, 0.0);
    glMatrixMode(GL_PROJECTION);
```

```
    gluOrtho2D(0.0, 900.0, 0.0, 800.0);  
}
```

```
void mydisplay() {  
    glClear(GL_COLOR_BUFFER_BIT);  
    //blue behind boat  
    glBegin(GL_POLYGON);  
    glColor3f(0.4, 0.7, 0.9);  
    glVertex2i(0, 0);  
    glVertex2i(1100, 0);  
    glVertex2i(1100, 300);  
    glVertex2i(0, 100);  
    glEnd();
```

```
    //green behind house  
    glBegin(GL_POLYGON);  
    glColor3f(0.5, 0.8, 0.2);  
    glVertex2i(0, 100);  
    glVertex2i(1100, 300);  
    glVertex2i(1100, 380);  
    glVertex2i(0, 380);  
    glEnd();  
    //mountains  
    glBegin(GL_TRIANGLES);  
    glColor3f(0.82, 0.55, 0.28);
```

```
    glVertex2i(0, 380);  
    glVertex2i(100, 450);  
    glVertex2i(200, 380);
```

```
    glVertex2i(100, 380);  
    glVertex2i(200, 450);  
    glVertex2i(300, 380);
```

```
    glVertex2i(250, 380);  
    glVertex2i(350, 460);  
    glVertex2i(450, 380);
```

```
    glVertex2i(400, 380);  
    glVertex2i(500, 470);
```

```

glVertex2i(600, 380);

glVertex2i(550, 380);
glVertex2i(650, 450);
glVertex2i(750, 380);

glVertex2i(700, 380);
glVertex2i(800, 480);
glVertex2i(900, 380);

glVertex2i(850, 380);
glVertex2i(950, 420);
glVertex2i(1050, 380);

glEnd();

//boat
glBegin(GL_POLYGON);
glColor3f(0.0, 0.0, 0.8);
glVertex2i(350, 40);
glVertex2i(550, 40);
glVertex2i(600, 90);
glVertex2i(300, 90);
glEnd();

//sticks
glBegin(GL_POLYGON);
glColor3f(0.3, 0.3, 0.3);
glVertex2i(455, 90);
glVertex2i(465, 90);
glVertex2i(465, 160);
glVertex2i(455, 160);
glEnd();

glBegin(GL_POLYGON);
glColor3f(0.3, 0.3, 0.3);
glVertex2i(475, 90);
glVertex2i(485, 90);
glVertex2i(485, 160);
glVertex2i(475, 160);

```

```

glEnd();

//flag
glBegin(GL_TRIANGLES);
glColor3f(0.8, 0.0, 0.0);
glVertex2i(450, 160);
glVertex2i(300, 95);
glVertex2i(450, 95);
glEnd();

//house box
glBegin(GL_POLYGON);
glColor3f(0.0, 0.6, 0.3);
glVertex2i(500, 220);
glVertex2i(650, 220);
glVertex2i(650, 340);
glVertex2i(500, 340);
glEnd();
//roof
glBegin(GL_POLYGON);
glColor3f(0.5, 0.2, 0.0);
glVertex2i(450, 340);
glVertex2i(700, 340);
glVertex2i(650, 400);
glVertex2i(500, 400);
glEnd();

//door
glBegin(GL_POLYGON);
glColor3f(0.0, 0.4, 0.3);
glVertex2i(560, 220);
glVertex2i(590, 220);
glVertex2i(590, 310);
glVertex2i(560, 310);
glEnd();

// Left Window
glBegin(GL_POLYGON);
glColor3f(0.0, 0.4, 0.3);
glVertex2i(520, 260);

```

```
glVertex2i(550, 260);
glVertex2i(550, 300);
glVertex2i(520, 300);
glEnd();

// Right Window
glBegin(GL_POLYGON);
glColor3f(0.0, 0.4, 0.3);
glVertex2i(600, 260);
glVertex2i(630, 260);
glVertex2i(630, 300);
glVertex2i(600, 300);
glEnd();

// House Box2
glBegin(GL_POLYGON);
glColor3f(0.8, 0.0, 0.0);
glVertex2i(100, 270);
glVertex2i(250, 270);
glVertex2i(250, 390);
glVertex2i(100, 390);
glEnd();

//roof
glBegin(GL_POLYGON);
glColor3f(0.5, 0.2, 0.0);
glVertex2i(80, 390);
glVertex2i(270, 390);
glVertex2i(250, 430);
glVertex2i(100, 430);
glEnd();

// Door
glBegin(GL_POLYGON);
glColor3f(0.0, 0.4, 0.3);
glVertex2i(160, 270);
glVertex2i(190, 270);
glVertex2i(190, 330);
glVertex2i(160, 330);
glEnd();
```

```
// Left Window
glBegin(GL_POLYGON);
glColor3f(0.0, 0.4, 0.3);
glVertex2i(120, 320);
glVertex2i(150, 320);
glVertex2i(150, 350);
glVertex2i(120, 350);
glEnd();
```

```
// Right Window
glBegin(GL_POLYGON);
glColor3f(0.0, 0.4, 0.3);
glVertex2i(200, 320);
glVertex2i(230, 320);
glVertex2i(230, 350);
glVertex2i(200, 350);
glEnd();
```

```
// House Box3
glBegin(GL_POLYGON);
glColor3f(0.8, 0.0, 0.0);
glVertex2i(-40, 250);
glVertex2i(35, 250);
glVertex2i(35, 370);
glVertex2i(-40, 370);
glEnd();
```

```
//roof
glBegin(GL_POLYGON);
glColor3f(0.5, 0.2, 0.0);
glVertex2i(-40, 370);
glVertex2i(55, 370);
glVertex2i(35, 400);
glVertex2i(-40, 400);
glEnd();
```

```
// Door
glBegin(GL_POLYGON);
glColor3f(0.3, 0.2, 0.1);
```



```

glVertex2i(-30, 250);
glVertex2i(-5, 250);
glVertex2i(-5, 300);
glVertex2i(-30, 300);
glEnd();

// Right Window
glBegin(GL_POLYGON);
glColor3f(0.0, 0.4, 0.3);
glVertex2i(10, 310);
glVertex2i(30, 310);
glVertex2i(30, 330);
glVertex2i(10, 330);
glEnd();

//first tree
glBegin(GL_POLYGON);
glColor3f(0.5, 0.2, 0.0);
glVertex2i(400, 200);
glVertex2i(420, 200);
glVertex2i(420, 300);
glVertex2i(400, 300);
glEnd();

glBegin(GL_TRIANGLES);
glColor3f(0.0, 0.6, 0.0);
glVertex2i(340, 300);
glVertex2i(410, 370);
glVertex2i(480, 300);
glEnd();

glBegin(GL_TRIANGLES);
glVertex2i(350, 340);
glVertex2i(410, 400);
glVertex2i(470, 340);
glEnd();

glBegin(GL_TRIANGLES);
glVertex2i(360, 375);
glVertex2i(410, 430);

```

```

glVertex2i(460, 375);
glEnd();

//second tree
glBegin(GL_POLYGON);
glColor3f(0.5, 0.2, 0.0);
glVertex2i(290, 200);
glVertex2i(310, 200);
glVertex2i(310, 280);
glVertex2i(290, 280);
glEnd();

glBegin(GL_TRIANGLES);
glColor3f(0.0, 0.6, 0.0);
glVertex2i(250, 275);
glVertex2i(300, 330);
glVertex2i(350, 275);
glEnd();

glBegin(GL_TRIANGLES);
glVertex2i(255, 310);
glVertex2i(300, 360);
glVertex2i(345, 310);
glEnd();

glBegin(GL_TRIANGLES);
glVertex2i(260, 340);
glVertex2i(300, 390);
glVertex2i(340, 340);
glEnd();

//third tree
glBegin(GL_POLYGON);
glColor3f(0.5, 0.2, 0.0);
glVertex2i(160, 200);
glVertex2i(175, 200);
glVertex2i(175, 260);
glVertex2i(160, 260);
glEnd();

glBegin(GL_TRIANGLES);

```

```
glColor3f(0.0, 0.6, 0.0);  
glVertex2i(130, 260);  
glVertex2i(168, 300);  
glVertex2i(205, 260);  
glEnd();
```

```
glBegin(GL_TRIANGLES);  
glVertex2i(135, 285);  
glVertex2i(168, 320);  
glVertex2i(200, 285);  
glEnd();
```

```
glBegin(GL_TRIANGLES);  
glVertex2i(140, 310);  
glVertex2i(168, 340);  
glVertex2i(195, 310);  
glEnd();
```

```
//first tree  
glBegin(GL_POLYGON);  
glColor3f(0.5, 0.2, 0.0);  
glVertex2i(745, 230);  
glVertex2i(775, 230);  
glVertex2i(775, 320);  
glVertex2i(745, 320);  
glEnd();
```

```
glBegin(GL_TRIANGLES);  
glColor3f(0.0, 0.6, 0.0);  
glVertex2i(720, 320);  
glVertex2i(760, 400);  
glVertex2i(800, 320);  
glEnd();
```

```
//second tree  
glBegin(GL_POLYGON);  
glColor3f(0.5, 0.2, 0.0);  
// Trunk (Taller)  
glVertex2i(845, 200);  
glVertex2i(875, 200);  
glVertex2i(875, 300); // Increased height
```

```

    glVertex2i(845, 300);
    glEnd();

    glBegin(GL_TRIANGLES);
    glColor3f(0.0, 0.6, 0.0);
    glVertex2i(820, 300);
    glVertex2i(860, 380);
    glVertex2i(900, 300);
    glEnd();

    // Cloud 1
    glBegin(GL_POLYGON);
    glColor3f(1.0, 1.0, 1.0);
    glVertex2i(200, 620);
    glVertex2i(300, 620);
    glVertex2i(300, 670);
    glVertex2i(200, 670);
    glEnd();

    // Sun
    glBegin(GL_TRIANGLES);
    glColor3f(1.0, 1.0, 0.0);
    glVertex2i(460, 600);
    glVertex2i(380, 530);
    glVertex2i(540, 530);
    glEnd();

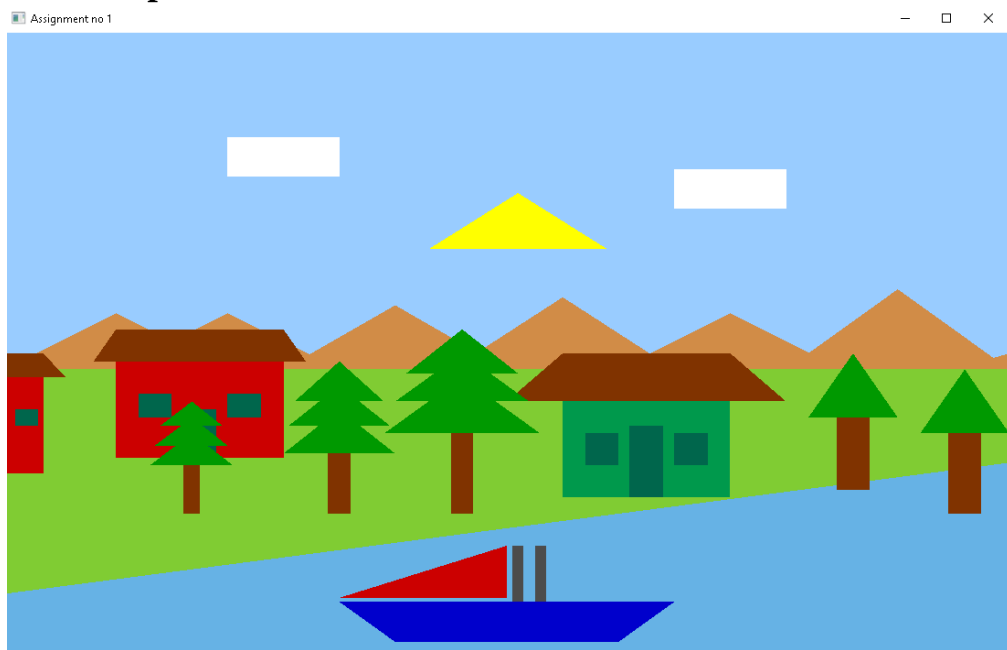
    // Cloud 2
    glBegin(GL_POLYGON);
    glColor3f(1.0, 1.0, 1.0);
    glVertex2i(600, 580);
    glVertex2i(700, 580);
    glVertex2i(700, 630);
    glVertex2i(600, 630);
    glEnd();

    glFlush();
}

```

```
int main(int argc, char* argv[]) {  
    glutInit(&argc, argv);  
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);  
    glutInitWindowSize(1100, 700);  
    glutCreateWindow("Assignment no 1");  
    glutDisplayFunc(mydisplay);  
    MyInit();  
    glutMainLoop();  
}
```

1.2 Output



1.3 Description

This OpenGL scene depicts a simple landscape with a blue sky, white clouds, a yellow sun, and brown mountains in the background. The green land has two houses, trees, and a river in the foreground. A boat with a red cabin is floating on the river. The scene is created using basic geometric shapes and bright colors.
